

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	29 September 2026
Team ID	SWTID-2026-7735
Project Name	AI Blog Nest API
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration & Authentication	Email/password registration, credential validation, secure password storage and login/token issuance.
FR-2	Role-Based Access Control	Authenticate requests and restrict protected routes/core settings according to user role.
FR-3	Blog CRUD Management	Create, read, update and delete blog posts with content, status, categories/tags and asset references.
FR-4	AI Blog Generation	Generate structured full-length blog posts from high-level themes and contextual inputs using an external LLM.
FR-5	AI Content Summarization	Generate concise abstracts and metadata summaries from extensive article content.
FR-6	Categorization & Tagging	Create, assign, filter and query tags/categories for improved discoverability.
FR-7	Comment Moderation	Manage comment moderation flags including approved, pending and spam through protected routes.
FR-8	Asset Assignment	Associate supported media assets with blog posts while maintaining separate asset storage.
FR-9	Input Validation & Sanitization	Validate and sanitize request parameters to reduce XSS and NoSQL parameter-manipulation risks.
FR-10	Search & Filtering	Support API-level querying by category, tag, status and other available metadata.

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	Consistent endpoints and response structures with clear validation/error messages.
NFR-2	Security	JWT authentication, role checks, password hashing, sanitization, security headers and rate limiting.
NFR-3	Reliability	Controlled error handling for database and AI failures, with safe

		processing and logging.
NFR-4	Performance	Indexed queries, pagination, bounded payloads and avoidance of unnecessary AI calls.
NFR-5	Availability	Cloud-ready stateless API, managed database, health checks and restart/recovery support.
NFR-6	Scalability	API instances can scale horizontally while database and external AI services scale independently.

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Main Purpose: AI Blog Nest API simplifies blog creation and management using AI-powered content generation and summarization while providing a secure backend for creating, managing and organizing blog content.